POLY PMNA · DIPLOMA CURRICULUM REV2021

Course 5002 | Semester 5 | Project | Shared handbook

Project Management and Software Engineering

Connect project-management discipline with software-engineering practice: requirements, estimation, design, testing, release, maintenance and responsible teamwork.

Course code	**5002**
Revision	**REV2021**
Semester	**Semester 5**
Type	**Project**
Catalogue scope	**10 catalogue programme assignments**

COURSE ORIENTATION

How to use this handbook

This page is a structured learning aid for the shared catalogue record. Study the concept, complete the applied activity, retain the evidence, and compare the result with the latest approved programme instructions. Where the official syllabus differs, the official record takes precedence.

Source and verification note: The course identity, code, semester and subject type are taken from the tracked POLY PMNA REV2021 catalogue, which indexes the official SITTTR scheme. The official course-content reference is SITTTR course 5002; the scheme index is SITTTR REV2021 diploma syllabus. The direct subject-content page was not machine-readable or returned an unavailable-file response during preparation, so module headings and explanatory teaching material on this page are an original study-handbook structure, not claimed verbatim SITTTR wording. Confirm current hours, marks, credits, outcomes and assessment against the latest official programme record before examination use.

Learning outcomes

- Explain the central concepts and vocabulary of the subject using clear technical language.
- Apply the subject method to a diploma-level problem, project, laboratory or workplace situation.
- Create evidence such as a plan, diagram, record, test, report, presentation or reflection.
- Identify assumptions, limitations and common errors, then improve the work using feedback.

Shared-record context

This code is assigned to 10 catalogue programme assignments in the tracked catalogue. The page therefore focuses on common learning practice and avoids claiming programme-specific technical details that were not verified.

Study rule: Do not treat the author-created examples as official examination questions or official mark distribution.

STUDY MAP

Modules and evidence

Author-created study map; verify official hours and marks			
Module	**Heading**	**Purpose**	**Evidence to retain**
1	Projects, stakeholders and life cycles	Use a shared vocabulary for projects and select a life-cycle approach that matches uncertainty and delivery needs.	Classify a diploma project’s tasks, stakeholders, dependencies and milestones.
2	Requirements engineering and estimation	Transform stakeholder needs into traceable requirements and a plan that can be monitored.	Write five testable requirements and estimate them using a small work-breakdown structure.
3	Risk, quality and software design	Control uncertainty while making design decisions that keep a product understandable and maintainable.	Create a risk register and critique two alternative designs for the same feature.
4	Implementation, verification and validation	Use disciplined implementation and testing to find defects before release.	Build a requirements-to-test matrix and explain why one test cannot prove overall quality.
5	Release, maintenance and case study	Plan delivery and maintenance after the first working version is complete.	Prepare a release checklist, maintenance log and post-project review for a small software product.

MODULE 1

Projects, stakeholders and life cycles

Purpose-led study **Apply + reflect**

Use a shared vocabulary for projects and select a life-cycle approach that matches uncertainty and delivery needs.

Core topics−

- Project, programme, portfolio and deliverable
- Stakeholders, milestones, dependencies and issues
- Predictive, iterative, incremental and hybrid life cycles
- Definition of done and acceptance criteria

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Classify a diploma project’s tasks, stakeholders, dependencies and milestones.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 2

Requirements engineering and estimation

Purpose-led study | Apply + reflect

Transform stakeholder needs into traceable requirements and a plan that can be monitored.

Core topics−

- Interviews, observation, use cases and user stories
- Functional and non-functional requirements
- Prioritisation and traceability

- Work breakdown, effort, duration, resources and cost

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Write five testable requirements and estimate them using a small work-breakdown structure.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
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Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 3

Risk, quality and software design

Purpose-led study | Apply + reflect

Control uncertainty while making design decisions that keep a product understandable and maintainable.

Core topics−

- Risk register and probability-impact analysis
- Quality criteria, reviews and audits
- Modularity, abstraction, cohesion and coupling
- Interfaces, architecture diagrams and trade-offs

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Create a risk register and critique two alternative designs for the same feature.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.

3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 4

Implementation, verification and validation

Purpose-led study | Apply + reflect

Use disciplined implementation and testing to find defects before release.

Core topics−

- Coding standards, version control and review
- Unit, integration, system and acceptance testing
- Regression, usability, performance and security checks
- Defect reports and evidence-based closure

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Build a requirements-to-test matrix and explain why one test cannot prove overall quality.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.

- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

MODULE 5

Release, maintenance and case study

Purpose-led study | Apply + reflect

Plan delivery and maintenance after the first working version is complete.

Core topics−

- Configuration and release management
- Issue tracking, change control and versioning
- Corrective, adaptive and perfective maintenance
- End-to-end case study from requirements to release

Study method

Read the concept, connect it to a diploma-level situation, make a labelled record or diagram, and check the result against the stated outcome. Keep programme-specific technical decisions within the approved project, laboratory or workplace context.

Applied activity

Prepare a release checklist, maintenance log and post-project review for a small software product.

Evidence: Keep the completed activity, a short reflection and any source or test record in your course folder.

Common mistakes and corrections−

- Starting with implementation before defining the problem, inputs, outputs or acceptance criteria.
- Making a claim without a source, measurement, test or documented observation.
- Using a template mechanically instead of adapting it to the approved programme context.

Correction: state the assumption, choose evidence, perform the check, and record the decision.

Self-check−

1. Explain the main idea of this module in your own words.
2. Apply it to a small technical or workplace scenario.
3. Identify one likely failure and describe the evidence that would diagnose it.

ASSESSMENT PREPARATION

Evidence, revision and quality gates

Before submission or examination

1. Confirm the current SITTTR/SBTE title, hours, credits, outcomes and assessment.
2. Define the problem or prompt and state assumptions.
3. Show the method, calculation, algorithm, project record or communication structure.
4. Attach test, source, supervisor, workplace or reflection evidence as applicable.
5. Review clarity, safety, ethics, citations, originality and accessibility.

Revision checklist

- Can I define the main terms without copying?
- Can I apply the method to a new situation?

- Can I explain one common error and its correction?
- Can I show evidence for each claimed outcome?
- Can I state the limitations and the next improvement?

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Evidence record
Subject: 5002 — Project Management and Software Engineering
Task or question:
Assumption or requirement:
Method / design / procedure:
Result or artefact:
Test / source / supervisor evidence:
Reflection and next action:
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SELF-CHECK AND EXAMINATION PRACTICE

Question bank

Recall question

Define two important terms from Project Management and Software Engineering and distinguish them.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Explain question

Explain why the method in this handbook matters for a diploma student.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Apply question

Apply one module method to a realistic technical, project or workplace situation.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Evaluate question

Identify a weak approach, state the evidence needed, and propose a defensible improvement.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

Create question

Design a small artefact, plan, test, report section or presentation that demonstrates the subject outcomes.
Answer guidance: define the terms, state assumptions, show the method, use a specific example, and cite or retain evidence where appropriate.

REFERENCES AND GLOSSARY

Reference trail

1. Official SITTTR REV2021 diploma syllabus catalogue.
2. Official SITTTR course-content reference for code 5002.
3. POLY PMNA Study Hub, for the current revision index and related lesson pages.

Glossary

Terms used in this handbook	
Term	**Working meaning**
Outcome	A capability the learner should demonstrate, not just a topic read.
Evidence	A record, artefact, result, source, test or review that supports a claim.
Acceptance criterion	A condition used to decide whether a requirement or deliverable is satisfactory.
Reflection	A brief evidence-based account of what worked, what did not and what should change.

Prepared as a POLY PMNA study handbook. Always follow the latest official SITTTR/SBTE instructions for the programme and examination session.